

Going Global with Agile Service Networks

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Abstract—ASNs are emergent networks of service-based applications (nodes) which collaborate through agile (i.e. adaptable) transactions. GSE comprises the management of project teams distanced in both space and time, collaborating in the same development effort. The GSE condition poses challenges both technical (e.g. geolocalization of resources, information continuity between timezones, etc.) and social (e.g. collaboration between different cultures, fear of competition, etc.). ASNs can be used to build an adaptable social network (ASN_{GSE}) supporting the collaborations (edges of ASN_{GSE}) of GSE teams (nodes of ASN_{GSE}).

Keywords—Agile Service Networks; Global Software Engineering; SOA; Social Networks; Organizational Structures; Cloud Computing

I. INTRODUCTION

GSE is a software engineering strategy in which organizations try to achieve round-the-clock productivity by distributing production along different continents and timezones [10], [11], [19]. GSE is a business decision. As a consequence of this decision, teams' time and space distance pose challenges both social and technical. Social challenges, such as cultural barriers, language difference, social class, unawareness of others limit the degree to which teams involved in GSE are able to share information effectively and collaborate efficiently. Technical challenges, such as space distance, information discontinuity between timezones, geolocalization of development resources hinder the way productivity can be coordinated on a global scale. These challenges inhibit teams' communication and collaboration to a point where project failure risk increases [11], [8], [19]. Major causes for delays and cost explosion in GSE projects are miscommunication on project requirements, deadlines and architecture [4]. To exemplify these issues consider the following simple (yet common) scenario [5], [9]:

“An unavoidable technical constraint discovered by a programmer causes design to be reworked, designers negotiate an adjustment in requirements with system analysts and modify their designs appropriately. Subsequently, this technical constraint is raised repeatedly by other programmers who were not informed of the design changes (or are using designs that may have been unrelated to the adjusted

design, but related to the affected requirement). This chaotic disruption wastes time and effort and causes unnecessary aggravation to the development team.”

GSE technical difficulties should be supported through socially-enabled technologies for the above challenges to be overcome [11]. These technologies would be able to avert chaos by automatically linking together the people involved, increasing their awareness. ASNs are emerging networks of collaborative service applications (nodes) which interoperate through agile transactions (edges), i.e. transactions which react to context change dynamically adapting themselves [2]. ASNs stem from the business decision of corporate collaborations to be formed globally, in pursuit of business gain for the nodes involved (i.e. every corporate partner) [6]. A fundamental similarity can be identified between GSE and ASNs. Both stem from business decisions. Moreover, a crucial complementarity exists between them. On one side, GSE needs dynamism among nodes (development teams) and their collaboration towards business gain (timely delivery). On the other, ASNs are supporting dynamic collaborations among nodes which are teaming up to increase business gain. These relations led us to consider three potentials of ASNs: (a) Collaborativeness as well as context adaptability make ASNs ideal to develop flexible social-networking tools for GSE; (b) being service-based, ASNs can be used in the cloud, using its services to further support GSE; (c) ASNs' emergence makes them ideal to build dynamic communication tools against distance in time and space. Based on these considerations, we argue that GSE challenges can be overcome through an ASN-based social network (ASN_{GSE}) providing agility of communications and collaborations (edges of ASN_{GSE}) to globally located IT professionals (nodes of ASN_{GSE}). Global professionals can be represented as nodes in an agile (i.e. adaptable and emergent) organizational social network to deliver the final product, just-in-time and sufficiently-good. The next section provides detail of the research design and challenges for this proposal. Section IV provides the main contributions we plan to obtain; section II recaps related work. Finally, section V concludes this paper.

II. RELATED WORK

We find our efforts most similar to [12], [16], [11] and [7], both in terms of intents and motivation.

In [12] authors represent the industrial/business perspective on digital formations and global development; [16] and [11] both provide empirical studies in GSE that stress the presence of social factors which shape the outcome of global development projects. Finally, [7], [15] provides exploration of social networking technologies as service networks.

In [12] authors examine political, social, cultural and economic implication of the digital revolution. They explore their influences in modern organizational activities such as business planning, project management, etc. They conclude that digital formations (i.e. new social structures which emerge in digitally-supported societies) should be considered a frontier for further research and development efforts. From the premises and conclusions of this work we draw motivation for our effort, since we try to research into organizational social structures (a concept from sociology) and use these results to shape a socially-enabled technology.

In [16] authors recall delay as a typical problem to be tackled in GSE. They explore this problem, its implication in GSE as well as its actuality. Empirical data from IBM leads the authors to observe that indeed distance and (mishandled) collaboration are crisis causes in GSE. The authors use social-network analysis to obtain data on collaboration. Our (empirical) plan is similar to such research. Our approach is top-down since we try to build a social-network and validate its functions in real-life GSE scenarios, rather than bottom-up by exploring GSE industrial scenarios to reverse engineer requirements. The same observations can be made for [11], since the authors operate in a manner similar to [16]. More in particular in [11], authors provide us motivations to invest in supporting the GSE OSS since the authors conclude that social networks can play a serious role in increasing the mutual awareness of professionals involved in GSE.

Finally, in [7], quoting the authors, “[the aim is] a framework integrating Web 2.0 social aspects to automatically enrich Web services with semantic meanings based on community consensus”. In much the same way, we propose to use both technical and social technologies to enrich the current collaboration possibilities among GSE professionals, through an agile service *social* network. In earlier works such as [15], the same idea was argued as supporting collaboration needs in current markets.

III. RESEARCH DESIGN

This section provides two elements: (a) the problem we want to tackle; (b) the solution we advocate.

A. Problem Statement

Distances in time and space make it impossible for GSE teams to communicate and coordinate their effort in an efficient manner [11]. Technical challenges, such as information

discontinuity between timezones, limit the advantages of GSE. Social challenges, such as language difference and social class difference, inhibit collaborativeness among GSE teams, increasing its chances of failure [19].

How can ASNs be used to support GSE?

B. Our Proposed Solution

The solution we advocate is an ASN-based social network (ASN_{GSE}) with the following characteristics:

- 1) *Agile context awareness.* ASNs are able to detect changes in the context and dynamically support different scenarios as needed. In GSE for instance, round-the-clock productivity could be supported by dynamically allocating collaborations between teams, by modeling each developer as a set of skills and allowing for their seamless (re-)allocation based on their timezone, location and needs. Also, seamless hand-off of relevant informations between two contiguous timezones could be used to ease the coordination of sequential or dependent work packages.
- 2) *Deployed in the cloud.* Cloud computing, has potentials which fit with GSE needs [14], [20]. For instance, GSE resources rendered available in the cloud allow for rapid resource location and access on a global scale. Also, communication and information continuity between timezones may be requested as needed.
- 3) *Satisfying GSE social requirements.* GSE teams together create an Organizational Social Structure (OSS) [17], part of a global corporation. Social interactions in OSSs depend on personal- or corporate-specific practices, which include work habits, methods, technologies to support cooperation, etc. In GSE, for instance, supporting social interactions among developers from different companies and cultures, would require letting them use own tools, languages and own methods seamlessly. ASNs can help in doing that through adaptable creation of service compositions or transparent information proxying (i.e. providing seamless switching between answering nodes in case of difficulty). Another example could be using ASNs to autonomously try and assemble a common tool-workbench for all teams. Lastly, an ASN to support GSD could compute the work allocation to teams, using (up-to-date and context-aware information) on project requirements, time constraints, etc.
- 4) *Project-centric as well as People-centric.* Enterprise Social Networking technologies already exist which could potentially represent (and supporting) the social network of an entire corporation. What is still missing is the dynamic / automatic adjustment of its granularity, to support the global software development project (against its changing context) as well as the people involved (e.g. technicians, developers, managers, etc.). Moreover, none of these technologies provide flexible

and adaptable collaboration channels (e.g. adaptable status-tracking, always-on reachability of key roles, worldwide project chatter, etc.) among professionals collaborating in the same GSE effort.

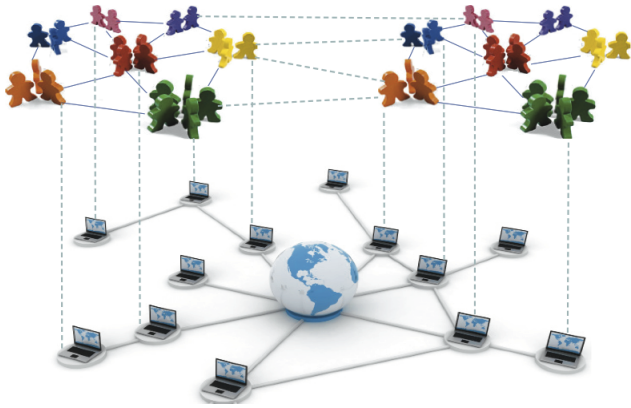


Figure 1. ASNs for GSD

Figure 1 tries to visualize the concept we have in mind. In our vision different globally-distributed teams (upper part) can be linked with devices (e.g. computers, in the lower part). The network links professionals (below) but brings them together into an emergent social network (above) which supports their collaboration as well as their sense of participation to a project. This can support them in coordinating their effort or prevent disaster scenarios.

With respect to the example in section I, the solution we have in mind would make the global software engineering network, more “agile” in three dimensions: (a) awareness - every node would be proactively aware of the status and presence of all others; (b) emergence - ASNs could increase-decrease their functionality through cloud, adding and removing capabilities, devices, people and commodities as needed; (c) organization - our ASNs would support the OSS of the current GSE, adapting it if needed, thereby strengthening the organizational efficiency of GSE efforts, by virtue of Conway’s law [3].

C. Research Methods

To develop a solution with the mentioned characteristics, we plan the following actions:

- 1) A systematic Literature Review (SLR) into OSSs. This review provides the requirements, features and potentials for social structures present in literature. Mapping GSE on these results can give us the profile of the OSS for GSE, as well as a picture of its context.
- 2) Case-study research on ASNs. We have to understand the potentials and limits of the context-awareness and adaptation mechanisms behind ASNs. For this we plan to study cases from domains in which dynamic adaptation is of extreme importance (e.g. emergency

management). These studies can validate feasibility of ASN adaptation mechanisms.

- 3) Context-Modeling. To allow ASNs’ adaptability to the GSE context, this must be clearly developed and formalized. Case-studies, interviews or other empirical research in industries involved in GSE, can be used to validate the obtained context-model. From a validated context-model a prototype of the social network can be developed and proposed in industry to be validated through action research.
- 4) Feasibility studies. To understand ASNs’ feasibility in the cloud, a requirements-based feasibility study of ASNs in the cloud must be carried out.
- 5) Prototyping. A prototype of ASN to support the GSE OSS must be developed to evaluate its practical value against our objectives. The prototype should model and support (at least a portion of) the GSE OSS, as well as adapting to (at least a portion of) changes.
- 6) Validation through action research. We are already in contact with large industrial partners to validate the a working prototype, using standard 3-to-6-month cycles of action research, across a three-year period. A working prototype will be deployed and evaluated in real industrial scenarios.

So far (8 months of investigation) we have obtained two key results: (a) we have established the feasibility of ASNs in the cloud [21], [18]; (b) we have sketched the very first version of a context-model for GSE [23]. Currently, we are finishing a systematic survey into OSSs [17].

IV. MAIN CONTRIBUTIONS

The dissertation about this research is expected to offer the following key contributions: (a) a clear-cut definition of ASNs, stating its defining characteristics, operational mechanics as well as (certified) applicative grounds (e.g. cloud computing and emergency management through case study research, GSE, etc.); (b) a validated OSS profile and context-model for GSE including social structure requirements, so that ad-hoc support tools may be (further) developed for it; (c) an ASN-based social-network prototype to support collaboration in GSE. The prototype would be validated in practice through action research.

V. CONCLUSION

In this paper we outlined the proposal of an ASN-based social network (ASN_{GSE}). We elaborated this proposal into a research plan. We have pointed towards preliminary results which are encouraging us in exploring the research path further. Specifically, by analyzing GSE and matching its challenges with ASNs we concluded that indeed ASNs are supportive [21]. Moreover, by inspecting GSE literature we were able to draw an initial version of a GSE context-model [23]. In the future we plan to increase this context-model with social requirements from an SLR we are conducting.

Finally, we plan to develop a very initial version of a prototype for the proposed social-networking ASN for GSE, to initiate industrial action-research validation.

LIST OF MY PREVIOUS PUBLICATIONS

I was an active author and main contributor in [13], [22], [1], [18], [23], [21].

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REFERENCES

- [1] Marco Autili, Paolo Di Benedetto, Paola Inverardi, and Damien A. Tamburri. Towards self-evolving context-aware services. *ECEASST*, 11, 2008.
- [2] Noel Carroll, Eoin Whelan, and Ita Richardson. Applying social network analysis to discover service innovation within agile service networks. *Service Science*, 2:pp 225–244, November 2010.
- [3] M.E. Conway. How do committees invent. *Datamation*, 14(4):28–31, 1968.
- [4] Daniela Damian. Stakeholders in global requirements engineering: Lessons learned from practice. *IEEE Software*.
- [5] Heather Oppenheimer Daniela Damian, Filippo Lanubile. The international workshop on global software development. In *Proceedings of ICSE Conference 2003*, 2003.
- [6] Shoumen Datta, Bob Betts, Mark Dinning, Feryal Erhun, Tom Gibbs, Pinar Keskinocak, Hui Li, Mike Li, and Micah Samuels. Adaptive value networks. In Yoon S. Chang, Harris C. Makatsoris, and Howard D. Richards, editors, *Evolution of Supply Chain Management*, pages 3–67. Springer US, 2004.
- [7] Khaled El-Goarany, Iman Saleh, and Gregory Kulczycki. The social service network - web 2.0 can make semantic web services happen. In *Proceedings of the 2008 10th IEEE Conference on E-Commerce Technology and the Fifth IEEE Conference on Enterprise Computing, E-Commerce and E-Services*, pages 419–423, Washington, DC, USA, 2008. IEEE Computer Society.
- [8] The Standish Group. *Standish Chaos Report*. 1995-2009.
- [9] Elizabeth Hargreaves, Daniela Damian, Filippo Lanubile, and James Chisan. Global software development: building a research community. *SIGSOFT Softw. Eng. Notes*, 29:1–5, September 2004.
- [10] James D. Herbsleb. Global software engineering: The future of socio-technical coordination. In Lionel C. Briand and Alexander L. Wolf, editors, *FOSE*, pages 188–198, 2007.
- [11] James D. Herbsleb, Audris Mockus, Thomas A. Finholt, and Rebecca E. Grinter. An empirical study of global software development: distance and speed. In *ICSE '01: Proceedings of the 23rd International Conference on Software Engineering*, pages 81–90, Washington, DC, USA, 2001. IEEE Computer Society.
- [12] Robert Latham and Saskia Sassen, editors. *Digital formations: IT and new architectures in the global realm*. Princeton University Press, Princeton, NJ, 2005.
- [13] Ivano Malavolta, Henry Muccini, Patrizio Pelliccione, and Damian A. Tamburri. Providing architectural languages and tools interoperability through model transformation technologies. *IEEE Transactions on Software Engineering*, 36(1):119–140, January 2010.
- [14] Ivanka Menken and Gerard Blokdijs. *Cloud Computing Best Practice Guide: Strategies, Methods and Challenges To Managing Services in the Cloud*. Emereo Pty Ltd, London, UK, UK, 2009.
- [15] Alexander Mikroyannidis. Toward a social semantic web. *IEEE Computer*, 40(11):113–115, 2007.
- [16] Thanh Nguyen, Timo Wolf, and Daniela Damian. Global software development and delay: Does distance still matter? *Global Software Engineering, 2008. ICGSE 2008. IEEE International Conference on*, pages 45–54, Aug. 2008.
- [17] K. Ruikar, L. Koskela, and M. Sexton. Communities of practice in construction case study organisations: Questions and insights. *Construction Innovation*, 9(4):434–, 2009.
- [18] Maryam Razavian Christina Manteli Patricia Lago Elisabetta Di Nitto Sajid Hashmi, Viktor Clerc, Ita Richardson, and Damian A. Tamburri. Software as a service in the cloud to support global software development. In *5th International Workshop on Tool Support and Requirements Management in Distributed Projects (REMIDI'11)*. 6th International Conference on Global Software Engineering (ICGSE 2011).
- [19] Raghvinder Sangwan, Matthew Bass, Neel Mullick, Daniel J. Paulish, and Juergen Kazmeier. *Global Software Development Handbook (Auerbach Series on Applied Software Engineering Series)*. Auerbach Publications, Boston, MA, USA, 2006.
- [20] Stefan Tai, Jens Nimis, Alexander Lenk, and Markus Klems. Cloud service engineering. In Jeff Kramer, Judith Bishop, Premkumar T. Devanbu, and Sebastin Uchitel, editors, *ICSE (2)*, pages 475–476. ACM, 2010.
- [21] Damian A. Tamburri and Patricia Lago. Satisfying cloud computing requirements through agile service networks. *Proceedings of SERVICES 2011*, 2011.
- [22] Damian A. Tamburri, Patricia Lago, and Henry Muccini. Leveraging software architectures through the iso/iec 42010 standard: A feasibility study. In *TEAR*, pages 71–85, 2010.
- [23] Lago P. Tamburri, D. A. Supporting communication and cooperation in global software development with agile service networks. *ECSCA 2011*, Springer:8 pages, 2011.